Grade 7 Mathematics Test: Fractions (Advanced)
Total Marks: 20

Instructions: Solve each question carefully. Show your work for full marks. Each question is labeled with its mark value.

Section A: Simplify the Following Fractions (4 Marks)

Simplify each fraction to its lowest form. Each question is worth 1 mark.

1. $-\frac{36}{48}$ (1 mark)
2. $\frac{45}{-75}$ (1 mark)
3. $-\frac{56}{72}$ (1 mark)
4. $\frac{64}{-96}$ (1 mark)

Section B: Evaluate the Following Expressions (6 Marks)

Evaluate each expression and express your answer in the simplest form. Each question is worth 2 marks.

5. $\left(\frac{2}{3}+\frac{5}{9}\right)\times\frac{3}{4}-\frac{1}{6}$ (2 marks)
6. $\frac{1}{5}\times\left(\frac{3}{8}-\frac{1}{4}\right)+\frac{7}{10}$ (2 marks)
7. $\left(\frac{4}{5}-\frac{2}{3}\right)\div\frac{1}{2}+\frac{3}{4}$ (2 marks)

Section C: Find the Multiplicative Inverse (4 Marks)

Find the multiplicative inverse of each mixed number or fraction. Each question is worth 1 mark.

8. $2\frac{1}{3}$ (1 mark)

9. $-3\frac{2}{5}$ (1 mark)

10. $\frac{4}{7}$ (1 mark)

11. $-\frac{5}{9}$ (1 mark)

Section D: Verify the Following (6 Marks)

Verify each equation by solving both sides and checking if they are equal. Each question is worth 2 marks.

12. $\left(\frac{3}{4}+\frac{2}{5}\right)\times\frac{1}{2}=\frac{7}{8}-\frac{1}{5}$ (2 marks)

13. $\left(\frac{5}{6}-\frac{1}{3}\right)\div\frac{2}{3}=\frac{1}{2}\times\frac{3}{4}-\frac{1}{8}$ (2 marks)

14. $\frac{2}{3}\times\left(\frac{1}{4}+\frac{3}{5}\right)-\frac{1}{6}=\left(\frac{4}{5}-\frac{1}{2}\right)\times\frac{3}{4}$ (2 marks)

Test Summary

- Section A: 4 questions, 1 mark each = 4 marks
- Section B: 3 questions, 2 marks each = 6 marks
- Section C: 4 questions, 1 mark each = 4 marks
- Section D: 3 questions, 2 marks each = 6 marks
- **Total:** 20 marks